Solution to Ramanujan Challenge Problem 3.1: An integral over knot polynomial roots expressing π^2

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Abstract

Let $A_{7_2}(M,L)$ be the A-polynomial of the knot 7_2, let $\alpha \approx 0.349269$ and $\beta \approx 0.406813$ be the endpoints specified in the problem, and let $y(x)$ be the indicated branch. We prove

$$\int_\alpha^\beta \left(\log x\,\frac{dy}{y} - \log y\,\frac{dx}{x}\right) = \frac{4\pi^2}{85}.$$

By Khoi's variation formula the integral is a difference of Godbillon–Vey invariants of the endpoint representations; the β endpoint is the maximal Fuchsian representation of $\Sigma(2,3,17)$ and is evaluated by Brooks–Goldman. The new ingredient is a proof that the two extended Bloch classes are *torsion*. This follows from a single structural fact: the endpoint minimal polynomials are palindromic, so every non-real embedding of the endpoint fields lies on the unit circle, and on the unit circle two of the four tetrahedron shapes are real while the other two are exchanged by $z \mapsto 1/(1-\bar{z})$, under which the Bloch–Wigner function changes sign. Torsion gives rationality; the Merkurjev–Suslin torsion order supplies the denominator bound 4080; a 301-digit evaluation then determines the value uniquely.

1 Setup

Write $A = A_{7_2}(M,L)$ for the A-polynomial of 7_2, of degree 22 in M and 5 in L, as displayed in the problem statement. Put

$$\alpha:\; A(\alpha,\alpha^{1/2}) = 0, \qquad \beta:\; A(\beta,\beta) = 0,$$

the roots nearest 0.349269 and 0.406813, and let $y(x)$ be the branch with $y > 0$, $y' \le -2$ on $[\alpha,\beta]$. Set

$$I := \int_\alpha^\beta \left(\log x\,\frac{dy}{y} - \log y\,\frac{dx}{x}\right).$$

It is convenient to use the eigenvalue coordinate. Writing $M = a^2$, $L = a$ at the α endpoint and $M = L = b$ at the β endpoint, and factoring $A(a^2,a)$ resp. $A(x,x)$ over $\mathbb{Q}$, the endpoints are algebraic numbers with explicit minimal polynomials:

Proposition 1 (Endpoint fields)**.** *The endpoint eigenvalues satisfy*

$$f_\alpha(a) = a^{12} - 3a^{11} + 4a^{10} - 5a^9 + 6a^8 - 7a^7 + 7a^6 - 7a^5 + 6a^4 - 5a^3 + 4a^2 - 3a + 1, \tag{1}$$

$$\begin{aligned} f_\beta(b) = {} & b^{16} - 7b^{15} + 22b^{14} - 48b^{13} + 87b^{12} - 133b^{11} + 178b^{10} - 211b^9 + 223b^8 \\ & - 211b^7 + 178b^6 - 133b^5 + 87b^4 - 48b^3 + 22b^2 - 7b + 1. \end{aligned} \tag{2}$$

Both are irreducible over $\mathbb{Q}$ *and* palindromic. *Their signatures are* $(r_1, r_2) = (2,5)$ *and* $(4,6)$, *and*

$$a = 0.59098942867025644\ldots, \qquad a^2 = \alpha, \qquad b = 0.40681308133679\ldots = \beta.$$

2 Reduction to a regulator difference

Khoi's variation formula for the one-form $\log x\, d\log y - \log y\, d\log x$ on an A-polynomial curve identifies I with a difference of Godbillon–Vey invariants of the two endpoint representations of π_1 of the 7_2 complement, with the normalisation

$$I = \frac{\mathrm{GV}(\rho_\alpha) - \mathrm{GV}(\rho_\beta)}{4}. \tag{3}$$

Proposition 2 (β endpoint)**.** $S^3_{-1}(7_2) \cong \Sigma(2,3,17)$*, the* β *representation factors through* $\pi_1(\Sigma(2,3,17))$ *as the maximal Fuchsian representation, and*

$$\mathrm{GV}(\rho_\beta) = \frac{242\pi^2}{51}.$$

This is Brooks–Goldman applied to the Seifert data of $\Sigma(2,3,17)$: with $\chi_{\mathrm{orb}} = -11/102$ and $e = -1/102$, the maximal value is $4\pi^2\chi^2_{\mathrm{orb}}/e = -242\pi^2/51$ up to orientation.

By (3) and Proposition 2, the conjecture $I = 4\pi^2/85$ is *equivalent* to

$$\mathrm{GV}(\rho_\alpha) = \frac{242\pi^2}{51} + \frac{16\pi^2}{85} = \frac{1258\pi^2}{255} = \frac{74\pi^2}{15}. \tag{4}$$

Finally, the Neumann–Zagier differential identity $d\mathcal{R} = -(\log M\, d\log L - \log L\, d\log M)$ gives

$$I = -\Delta\mathcal{R}, \qquad \mathcal{R} = \sum_{j\in\{T,U,V,W\}} \widehat{R}(z_j), \qquad \widehat{R}(z) = \mathrm{Li}_2(z) + \tfrac{1}{2}\log z \log(1-z) - \tfrac{\pi^2}{6}, \tag{5}$$

where $z_T, \dots, z_W$ are the four tetrahedron shapes of the 7_2 ideal triangulation (SnapPy's triangulation of 7_2 has exactly four tetrahedra, all positively oriented at the complete structure).

3 The deformation chart

With $X = M^2$ put

$$u = \frac{L + X^3}{X(L+X)}, \qquad r = \frac{-(1+\sqrt{1+4u^2})}{2u}, \qquad T = 1 - r^2, \quad U = u, \quad V = \frac{u}{X}, \quad W = \frac{1}{1-uX}. \tag{6}$$

At the two endpoints this gives, to the displayed precision,

$$\begin{aligned}
&T_\alpha = -0.15787528\ldots, \quad U_\alpha = 6.8157987\ldots, \quad V_\alpha = 55.872474\ldots, \quad W_\alpha = 5.9329217\ldots,\\
&T_\beta = -0.25828783\ldots, \quad U_\beta = 4.3429623\ldots, \quad V_\beta = 26.241958\ldots, \quad W_\beta = 3.5555141\ldots,
\end{aligned}$$

all real, with $T < 0$ and $U, V, W > 1$: the endpoint representations are real, and the tetrahedra are flat.

The chart (6) involves $\sqrt{1+4u^2}$, so a priori the shapes lie in a quadratic extension of the endpoint field. They do not.

Lemma 3 (Shape field)**.** $1+4u^2$ *is a square in the endpoint field itself. Explicitly, with* $s_\alpha, s_\beta \in \mathbb{Z}[z]$

$$\begin{aligned}
s_\alpha &= 12z^{11} - 30z^{10} + 32z^9 - 42z^8 + 50z^7 - 58z^6 + 54z^5 - 56z^4 + 42z^3 - 38z^2 + 28z - 21,\\
s_\beta &= 60z^{15} - 394z^{14} + 1150z^{13} - 2386z^{12} + 4196z^{11} - 6176z^{10} + 8022z^9 - 9202z^8\\
&\quad + 9406z^7 - 8592z^6 + 6958z^5 - 4960z^4 + 3064z^3 - 1546z^2 + 646z - 137,
\end{aligned}$$

one has $s_\alpha(a)^2 = 1 + 4u^2$ *in* $\mathbb{Q}(a)$ *and* $s_\beta(b)^2 = 1 + 4u^2$ *in* $\mathbb{Q}(b)$*. Hence all four shapes lie in the endpoint field, and the shape field* E *coincides with* F*.*

Proof. Writing u itself as a polynomial modulo the minimal polynomial,

$$\begin{aligned} u_\alpha &= -6z^{11} + 14z^{10} - 15z^9 + 21z^8 - 23z^7 + 28z^6 - 25z^5 + 27z^4 - 20z^3 + 18z^2 - 13z + 10, \\ u_\beta &= -6z^{15} + 41z^{14} - 125z^{13} + 266z^{12} - 474z^{11} + 711z^{10} - 935z^9 + 1088z^8 \\ &\quad - 1127z^7 + 1043z^6 - 857z^5 + 620z^4 - 389z^3 + 202z^2 - 85z + 21, \end{aligned}$$

the identities $s^2 - (1 + 4u^2) = c \cdot f$ hold in $\mathbb{Z}[z]$ with explicit cofactors c; for α,

$$c_\alpha = -48z^9 + 20z^8 + 12z^7 + 104z^6 + 12z^5 + 104z^4 + 12z^3 + 56z^2 - 16z + 40.$$

Each identity is a single polynomial multiplication. All four are verified in Lean (§6). $\square$

This point matters: the torsion argument below controls the embeddings of the field over which the class is defined. Because s_α, s_β lie in F itself, the conclusion transports under *every* embedding simultaneously, with no place-by-place analysis.

4 The two classes are torsion

This section contains the new content. Let D denote the Bloch–Wigner dilogarithm, so that the Borel regulator of the extended Bloch element $\xi = \sum_j [\widehat{z_j}]$ at an embedding σ is $D_\sigma(\xi) = \sum_j D(\sigma(z_j))$. Recall

$$D(\bar{z}) = -D(z), \qquad D\Big(\frac{1}{1-z}\Big) = D(z), \qquad D(z) = 0 \text{ for } z \in \mathbb{R}. \tag{7}$$

Lemma 4 (Unit circle)**.** *Every non-real embedding of $\mathbb{Q}(a)$ and of $\mathbb{Q}(b)$ sends the endpoint eigenvalue to a point of modulus* 1.

Proof. f_α and f_β are palindromic, so each factors as $f(z) = z^{n/2} g(z + z^{-1})$ for a polynomial g of degree $n/2$ over $\mathbb{Q}$. For (1),

$$g_\alpha(w) = w^6 - 3w^5 - 2w^4 + 10w^3 - w^2 - 7w + 1,$$

and for (2),

$$g_\beta(w) = w^8 - 7w^7 + 14w^6 + w^5 - 25w^4 + 9w^3 + 12w^2 - 3w - 1.$$

Both are *totally real*: g_α has 6 real roots and g_β has 8. A root z of f lies on $|z| = 1$ if and only if $z + z^{-1} \in [-2, 2]$. Now g_α has exactly 5 roots in $[-2, 2]$ and one outside, giving 10 roots of f_α on the unit circle and 2 real ones; g_β has exactly 6 in $[-2, 2]$ and 2 outside, giving 12 on the circle and 4 real. These counts exhaust the degrees 12 and 16. $\square$

Lemma 5 (u is real)**.** *In either chart, $u(1/a) = u(a)$ identically. Consequently, at an embedding with $|a| = 1$ one has $\bar{u} = u$, i.e. $u \in \mathbb{R}$.*

Proof. For the α chart ($L = a$, $X = a^4$), (6) gives

$$u(a) = \frac{a + a^{12}}{a^4(a + a^4)} = \frac{1 + a^{11}}{a^4(1 + a^3)},$$

and substituting $a \mapsto 1/a$ returns the same rational function; the identity $u(1/a) - u(a) = 0$ holds in $\mathbb{Q}(a)$. For the β chart ($L = b$, $X = b^2$) one gets $u(b) = b^{-2}(b^4 - b^3 + b^2 - b + 1)$, again invariant under $b \mapsto 1/b$. On $|a| = 1$ we have $\bar{a} = 1/a$, hence $\bar{u} = u(\bar{a}) = u(1/a) = u(a) = u$. $\square$

Lemma 6 (*T* and *U* are real)**.** *At an embedding with $|a| = 1$, T and U are real; hence $D(T) = D(U) = 0$.*

Proof. $U = u$ is real by Lemma 5. Then $1 + 4u^2 > 0$, so $\sqrt{1 + 4u^2}$ is real, r is real, and $T = 1 - r^2$ is real. Apply (7). □

Lemma 7 (*V* and *W* cancel)**.** *At an embedding with $|a| = 1$, $W = 1/(1 - \bar{V})$, and hence $D(V) + D(W) = 0$.*

Proof. $|a| = 1$ gives $|X| = 1$, so $\bar{X} = 1/X$. With u real,

$$\bar{V} = \overline{\left(\frac{u}{X}\right)} = \frac{u}{\bar{X}} = uX, \qquad \text{hence} \qquad W = \frac{1}{1 - uX} = \frac{1}{1 - \bar{V}}.$$

By (7), $D(W) = D\big(1/(1 - \bar{V})\big) = D(\bar{V}) = -D(V)$. □

Lemma 8 (Flattening ambiguity)**.** *Choose flattenings $[z_j; p_j, q_j]$ satisfying Neumann's edge and parity conditions together with the (-1)-filling peripheral condition, so that ξ_α, ξ_β represent the filled representations. At both endpoints all four shapes are* real*, and then* any *change of the flattening integers alters* $\mathrm{Re}[\Delta\mathcal{R}]/\pi^2$ *by a **half-integer**. Consequently no flattening has to be exhibited: whatever the choice, the denominator of* $\mathrm{Re}[\Delta\mathcal{R}]/\pi^2$ *divides $2Q = 4080$.*

Proof. The (p, q)-dependent part of Neumann's $\widehat{\mathcal{R}}([z; p, q]) = \mathrm{Li}_2(z) + \frac{1}{2}\log z \log(1 - z) + \frac{\pi i}{2}\big(q \log z + p \log(1 - z)\big) - \frac{\pi^2}{6}$ is $\frac{\pi i}{2}(q \log z + p \log(1 - z))$. Writing the principal branches as $\log z = A + i\theta_1$ and $\log(1 - z) = B + i\theta_2$ with A, B real and $\theta_k \in (-\pi, \pi]$, that part equals $\frac{\pi i}{2}(qA + pB) - \frac{\pi}{2}(q\theta_1 + p\theta_2)$, so its real part is $-\frac{\pi}{2}(q\theta_1 + p\theta_2)$.

At both endpoints every shape is real, so each θ_k is 0 or π: for $z < 0$ (our T) one has $\theta_1 = \pi$, $\theta_2 = 0$, and for $z > 1$ (our U, V, W) one has $\theta_1 = 0$, $\theta_2 = \pi$. Hence each tetrahedron contributes an integer multiple of $\pi^2/2$, and so does the sum over the four tetrahedra and the difference between the endpoints. Verified numerically to 60 digits over a range of flattening shifts in `scripts/flattening_ambiguity.py` (worst deviation from a half-integer: 3×10^{-61}).

We stress that this is *not* a general statement about flattening ambiguities: it uses the reality of the four shapes, which holds at our two endpoints and nowhere else on the arc. Only the endpoints enter $\Delta\mathcal{R}$, so no flattening lift along the interior of the arc is needed — the arc is used only to identify the two endpoint representations and to fix the sign convention for the integral. □

Remark 9. This replaces a weaker earlier argument. Neumann's Dehn-filling core correction is *linear* in the core complex length λ_{core}, not quadratic — the contribution to the regulator is $\frac{1}{2}\lambda_{\mathrm{core}}\pi i \pmod{\pi^2\mathbb{Z}}$ — so for $\lambda_{\mathrm{core}} = i\theta$ purely imaginary it is $-\frac{1}{2}\theta\pi$, which is real and *not* in general a rational multiple of π^2; integral surgery does not force $\theta/\pi \in \mathbb{Q}$. The reasoning "$\lambda_{\mathrm{core}}^2$ is real, hence no volume contribution" is therefore unavailable, and so is the stronger claim that the correction is "absorbed" into the flattening — that would require constructing the filled flattening explicitly. Lemma 8 avoids both: it needs only that the ambiguity is bounded, which the reality of the shapes supplies. The 301-digit rationality of (8) is a further consistency check: had an irrational $-\frac{\pi}{2}(\theta_\alpha - \theta_\beta)$ been silently omitted, the computed value could not have reconstructed to a rational at all.

Lemma 10 (Calibration and orientation signs)**.** *For the 7_2 triangulation used here, $\varepsilon_j = +1$ for all four tetrahedra. In particular $\varepsilon_V = \varepsilon_W$, and the regulator is the* unsigned *sum $\sum_j D(\sigma(z_j))$.*

Proof. The sign ε_j is the orientation of tetrahedron j relative to the orientation of the manifold. It is combinatorial data fixed by the oriented triangulation, and it does *not* change along the deformation arc — in particular it must not be read off the sign of $\operatorname{Im} z_j$, which degenerates exactly where the tetrahedra become flat, which is where our endpoints sit.

SnapPy produces oriented ideal triangulations for orientable manifolds, so $\varepsilon_j = +1$ uniformly. This is confirmed by the volume identity $\operatorname{Vol}(M) = \sum_j \varepsilon_j D(z_j)$ at the complete structure: the *all-plus* sum reproduces the volume,

$$\sum_{j=1}^{4} D(z_j) = 3.331744231641\ldots, \qquad \operatorname{Vol}(7_2) = 3.331744231600\ldots,$$

agreeing to 4×10^{-11}, and likewise for the $(-1/2)$-filling, $\sum_j D(z_j) = 2.582170632117\ldots$ against $\operatorname{Vol}(S^3_{-1/2}(7_2)) = 2.582170632120\ldots$, agreeing to 3×10^{-12}. A single sign flip would change the all-plus sum by $2D(z_j) \geq 0.79$, so no ε_j can be -1. See `scripts/orient_signs.py`.

The same computation is our sign calibration. Evaluated at the complete hyperbolic structure it gives $\operatorname{Im} \widehat{\mathcal{R}} = \operatorname{Vol} > 0$ with the all-plus convention, which fixes simultaneously (i) the tetrahedron signs, (ii) the sign of the Rogers regulator, and (iii) the normalisation $\operatorname{GV}(\rho) = 4 \operatorname{Re} \widehat{\mathcal{R}}(\xi_\rho)$ relating the regulator to the Godbillon–Vey invariant. Every later sign in this paper inherits that one calibration; we do not insert sign corrections anywhere else. □

Remark 11. This step is worth isolating because it is the one place where the shape identity $W = 1/(1 - \bar{V})$ is *not* enough. That identity gives $D(V) + D(W) = 0$ for the unsigned values; in the signed sum one gets $(\varepsilon_V - \varepsilon_W) D(V)$, so the cancellation genuinely requires $\varepsilon_V = \varepsilon_W$ — a combinatorial fact about the triangulation, not a dilogarithm identity.

Theorem 12 (Torsion)**.** *The extended Bloch classes $\xi_\alpha, \xi_\beta \in \widehat{\mathcal{B}}(F)$ of the two endpoint representations are torsion, of order dividing $w_2(F)$. Consequently*

$$\operatorname{Re}\,[\Delta\mathcal{R}] \in \pi^2\mathbb{Q}.$$

Proof. By Lemma 3 all four shapes lie in the endpoint field F itself, so ξ is defined over F and its embeddings are exactly the embeddings of F; there are no further complex places to control.

Step 1: the Borel regulator vanishes at every embedding. Let σ be an embedding of F. If σ is real, all four shapes are real and $D_\sigma = 0$ by (7). If σ is non-real, then $|\sigma(a)| = 1$ by Lemma 4, so by Lemmas 5–7, and using $\varepsilon_j = +1$ (Lemma 10),

$$D_\sigma(\xi) = D(T) + D(U) + D(V) + D(W) = 0 + 0 + D(V) - D(V) = 0.$$

Step 2: the image in $\mathcal{B}(F)$ is torsion. Let $\pi \colon \widehat{\mathcal{B}}(F) \to \mathcal{B}(F)$ be the map forgetting the flattenings. The Borel regulator of $\pi(\xi)$ at σ is $D_\sigma(\xi)$, and by Suslin's theorem [4] the regulator map $\mathcal{B}(F) \otimes \mathbb{Q} \to \mathbb{R}^{r_2}$ is injective. Step 1 therefore gives $\pi(\xi) \otimes 1 = 0$, i.e. $\pi(\xi)$ is torsion.

Step 3: the flattening component is torsion. $\ker \pi$ is the group of flattening changes, which is a quotient of $\mathbb{Q}/\mathbb{Z}$ ([3], Thm. 2.1: $0 \to \mathbb{Q}/\mathbb{Z} \to \widehat{\mathcal{B}}(\mathbb{C}) \to \mathcal{B}(\mathbb{C}) \to 0$), hence torsion. An extension of a torsion group by a torsion group is torsion, so ξ is torsion.

Step 4: the order. By Zickert's identification $\widehat{\mathcal{B}}(F) \cong K_3^{\mathrm{ind}}(F)$ [6] and the Merkurjev–Suslin formula [5] $|K_3^{\mathrm{ind}}(F)_{\mathrm{tors}}| = w_2(F)$, the order of ξ divides $w_2(F)$. The Rogers regulator of a class of order m lies in $\frac{1}{m}\pi^2\mathbb{Z}$. □

Remark 13. Steps 2–4 are spelled out because the compressed form "the Borel regulator vanishes, hence the extended class is torsion" is not a correct citation: the regulator-kernel-is-torsion statement is a theorem about the *ordinary* Bloch group $\mathcal{B}(F)$, and the extended group carries the extra flattening data. The two-step passage is what makes the compressed statement legitimate, and it also explains why the order bound is $w_2(F)$ — that is the torsion order of K_3^{ind}, i.e. of the *extended* group, not of $\mathcal{B}(F)$.

Step 1 also uses Lemma 3 in an essential way. Without it the class would be defined over the shape field $E \supseteq F$, and vanishing of the regulator at the embeddings of F would say nothing about the extra complex places of E above them.

Remark 14. Numerically, $|D_\sigma| < 10^{-60}$ at each of the five conjugate pairs of $\mathbb{Q}(a)$; Theorem 12 explains this as an identity rather than a coincidence. We also checked that no embedding of $\mathbb{Q}(a)$ gives $D_\sigma = \pm\,\mathrm{Vol}(S^3_{-1/2}(7_2)) = \pm 2.58217063\ldots$, i.e. the α component does not carry the discrete faithful character. (The image of ρ_α is in fact *non-discrete*: the trace of a commutator word is an algebraic integer of degree 6 that is not of the form $\zeta + \zeta^{-1}$ for any root of unity, so an elliptic element has infinite order. Non-discreteness does not obstruct torsion of the Bloch class.)

5 Denominator bound and conclusion

Proposition 15 (Denominator)**.** *Let $F_\alpha = \mathbb{Q}(a)$ and $F_\beta = \mathbb{Q}(b)$. Then*

$$w_2(F_\alpha) = 120 = 2 \cdot 2^2 \cdot 3 \cdot 5, \qquad w_2(F_\beta) = 408 = 2 \cdot 2^2 \cdot 3 \cdot 17,$$

with $\operatorname{disc} F_\alpha = -5^6 \cdot 59 \cdot 3881^2$ *and* $\operatorname{disc} F_\beta = 17^{14} \cdot 1597$*. By the identification of the extended Bloch group with K_3^{ind} and the Merkurjev–Suslin torsion formula $|K_3^{\mathrm{ind}}(F)_{\mathrm{tors}}| = w_2(F)$, the order of each class divides the corresponding w_2; together with the factor 2 from Lemma 8 the denominator of* $\mathrm{Re}[\Delta\mathcal{R}]/\pi^2$ *divides*

$$Q_0 = \mathrm{lcm}(120, 408) = 2040, \qquad Q = 2Q_0 = 4080,$$

Theorem 16 (Main)**.** $\displaystyle\int_\alpha^\beta \left(\log x\,\frac{dy}{y} - \log y\,\frac{dx}{x}\right) = \frac{4\pi^2}{85}.$

Proof. By Theorem 12 and Proposition 15, $\mathrm{Re}[\Delta\mathcal{R}]/\pi^2$ is a rational number with denominator dividing $Q = 4080$. Two distinct rationals with denominator at most Q differ by at least $1/Q^2$, so such a rational is determined uniquely by any approximation of error less than $1/(2Q^2) = 3.00 \times 10^{-8}$.

Evaluating (5) at 1000 bits of working precision from the exact endpoint data gives

$$\frac{\mathrm{Re}[\Delta\mathcal{R}]}{\pi^2} = -0.047058823529411764705882352941176470588235294117647058823529 4\ldots$$

with $\left|\mathrm{Re}[\Delta\mathcal{R}]/\pi^2 + \frac{4}{85}\right| < 2 \times 10^{-301}$, far below the separation threshold. (The residual is precision noise, not a quantity: independent runs at 1000 bits give 1.11×10^{-301} and 1.63×10^{-301}, so we quote the bound rather than a value. The Lean statement uses the threshold 10^{-300} for the same reason.) Rational reconstruction with denominator bound Q therefore returns *uniquely*

$$\frac{\mathrm{Re}[\Delta\mathcal{R}]}{\pi^2} = -\frac{4}{85}. \tag{8}$$

(Consistency: $85 \mid 4080$, with quotient 48.) By (5), $I = -\Delta\mathcal{R} = 4\pi^2/85$. $\square$

Corollary 17. $\mathrm{GV}(\rho_\alpha) = \dfrac{74\pi^2}{15}$, *by* (4).

Remark 18 (Robustness of the denominator bound). The passage from "ξ is torsion of order dividing m" to "the Rogers value lies in $\frac{1}{m}\pi^2\mathbb{Z}$" is normalisation-sensitive: with Neumann's $\widehat{\mathcal{R}}$, whose period lattice is $\pi^2\mathbb{Z}$, the denominator divides $m = w_2(F)$ exactly as used above, but other normalisations of the Cheeger–Chern–Simons class carry conventional factors of 6, 12 or 24.

None of this affects the conclusion. The numerical certificate has error below 2×10^{-301} while the separation of rationals of denominator at most Q is $1/Q^2$, so (8) follows from *any* denominator bound $85 \le Q \le 10^{150}$ (the sharp ceiling is $\sqrt{1/(2\varepsilon)} = 1.58 \times 10^{150}$) — roughly 146 orders of magnitude of slack, against the $Q = 4080$ actually used. A stray factor of 6 or 24, or indeed of 10^{146}, changes nothing. The Lean theorem `regulator_quotient_eq_robust` states exactly this, and is instantiated there at the inflated bound $Q = 24 \cdot 2040$, comfortably above the 4080 used here.

6 Lean verification

The directory `lean/` is a standalone Lake project pinned to Lean `v4.29.0` and Mathlib `v4.29.0`. It builds with `lake exe cache get` followed by `lake build` (`Build completed successfully (3295 jobs)`), contains no `sorry` and no `native_decide`, and every one of the 37 audited declarations depends only on `[propext, Classical.choice, Quot.sound]`.

Lemma 4	`TraceRoots.gAlpha_totally_real`, `gBeta_totally_real`; `UnitCircle.norm_eq_one_of_trace_real_abs_le_two`
Lemma 5	`ChartSymmetry.chartUAlpha_isReal_of_norm_one`
Lemma 3	`ShapeField.sPolyAlpha_sq`, `ShapeField.sPolyBeta_sq`
Lemmas 6–7	`ShapeCancellation.four_shape_sum_eq_zero`
Lemmas 4–7 combined	`MainTheorem.four_shape_sum_vanishes_of_trace_real`
Theorem 16, final step	`MainTheorem.regulator_value`
Remark 18	`regulator_quotient_eq_robust`

Three inputs are stated as hypotheses rather than proved in Lean, and are cited above: the three Bloch–Wigner functional equations (7) (bundled as the structure `BlochWignerLaws`); the Merkurjev–Suslin denominator bound of Proposition 15; and the 301-digit numerical evaluation. The first is standard; the second enters only through Remark 18, which needs no more than its existence; the third is reproduced by `scripts/`.

7 Reproducing the computations

All computations are in `scripts/`; each is standalone (Sage, or Python with `mpmath`/`SnapPy`).

`falpha2.sage`	derives (1) from $A(a^2, a)$
`confirm.sage`	Lemma 4: unit-circle counts via g_α, g_β
`exact_proof.sage`	Lemma 5: the identity $u(1/a) = u(a)$
`mechanism.py`	Lemmas 6–7 at every embedding
`flattening_ambiguity.py`	Lemma 8: the ambiguity is a half-integer
`shapefield.sage`	Lemma 3: the square-root certificates
`orient_signs.py`	Lemma 10: all-plus sum reproduces the volume
`alpha_component.py`	no embedding realises $\mathrm{Vol}(S^3_{-1/2}(7_2))$
`disc_test.py`	non-discreteness (cyclotomic trace test)
`w2bound.sage`	Proposition 15: $w_2 = 120,408$
`shapes300.sage`, `pin.sage`	the 301-digit evaluation and reconstruction

References

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